

Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_best_single_fixed_heuristic, phase8_full_solver_evolution, phase8_llm_static_selector, phase8_random_portfolio.

Run Metadata

- run_name: run_20260520_161413_a
- started_at_local: 2026-05-20 16:14:13
- finished_at_local: 2026-05-20 16:46:56
- duration_hhmm: 00:33
- duration_seconds: 1963.222
- seed_offset: 0
- replicate_label: a
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Run time (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	1667.927714	0.07226	0.007284	0.031222	0.0	0.0	True
phase8_random_portfolio	random_portfolio	none	0.195137	0.122877	536.870029	0.195137	0.037367	0.095493	0.0	0.0	True
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	1678.000271	0.072696	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLI B Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	1681.478314	0.072847	0.007284	0.031222	0.0	0.0	False
phase8_llm_static_selector	static_selector	none	0.229271	0.157011	258.545014	0.229271	0.043202	0.111754	0.0	0.62	True
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.214311	0.142051	431.301657	0.214311	0.024801	0.094621	0.0	2.0175	False
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.086887	0.014627	2831.301143	0.14749	0.000388	0.032256	0.22521	0.62	False
phase8_full_solver_evolution	full_solver	none	0.105441	0.033181	390.821043	0.105441	0.001007	0.039482	0.854268	0.76	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1667.927714 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.195137, selector regret 0.122877, runtime-adjusted gap 0.195137.
- Family transfer 0.037367 and combined transfer 0.095493.
- Runtime 536.870029 ms, runtime inflation -0.678122, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.072696.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 1678.000271 ms, runtime inflation 0.006039, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.072847.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1681.478314 ms, runtime inflation 0.008124, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.229271, selector regret 0.157011, runtime-adjusted gap 0.229271.
- Family transfer 0.043202 and combined transfer 0.111754.
- Runtime 258.545014 ms, runtime inflation -0.84499, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: nearest_neighbor_multistart:farthest_insertion_two_opt:farthest_insertion_two_opt:edge_preserving_restart:stag3:cand2:restart1.
- Controller rule summary: {"acceptance_bias": 0.01, "candidate_limit_offset": 2, "clustered_heuristic": "farthest_insertion_two_opt", "corridor_heuristic": "farthest_insertion_two_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive frozen-portfolio selector. Prefers farthest_insertion_two_opt as generalist; uses pruning for NN-trap risk, and restart-heavy edge_preserving_restart for stagnation-style recovery; corridor/grid/bottleneck mapped to farthest_ins", "failed_perturbation_threshold": 2, "grid_like_heuristic": "farthest_insertion_two_opt", "low_improvement_threshold": 0.05, "name": "tsp_frozen_portfolio_instance_rule_v1", "nearest_neighbor_trap_heuristic": "candidate_pruned_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 3, "temperature_scale": 1.05, "time_budget_trigger": 0.45, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.024801 and combined transfer 0.094621.
- Runtime 431.301657 ms, runtime inflation -0.741415, mean novelty 0.0, and complexity 2.0175.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cheapest_insertion_two_opt:farthest_insertion_two_opt:edge_preserving_restart:stag3:cand3:restart1.
- Controller rule summary: {"acceptance_bias": 0.004, "candidate_limit_offset": 3, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive controller selecting from frozen heuristic portfolio with bounded online scheduling switches.", "deterministic_randomization": {"mapping_fields": ["number of cities", "coordinate spread", "distance_mean_norm", "distance_cv", "mst_length_norm"], "method":

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"implicit_seed_from_instance_id"}, "failed_perturbation_threshold": 2, "grid_like_heuristic":
"farthest_insertion_two_opt", "low_improvement_threshold": 0.07, "name": "tsp_hh_deterministic_interpretable_v1",
"nearest_neighbor_trap_heuristic": "annealed_multi_start", "online_schedule_adjustments":
{"candidate_pruning_tuning": {"candidate_limit_offset": "candidate_limit_offset", "trigger": "if using
candidate-list-capable backbone (detected by heuristic name in {candidate_pruned_two_opt})"},
"late_phase_strategy": {"parameters": {"acceptance_bias": "acceptance_bias", "temperature_scale":
"temperature_scale", "then_heuristic": "annealed_multi_start", "trigger": "time budget used >= time_budget_trigger
AND accepted swaps is low"}, "stagnation_switch": {"perturbation_backoff": {"failed_perturbation_threshold":
"failed_perturbation_threshold", "offsets": {"restart_offset": "restart_offset"}}, "then_heuristic":
"edge_preserving_restart", "trigger": "stagnation length >= stagnation_threshold AND recent improvement rate <
low_improvement_threshold"}}, "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1,
"selection_policy": {"priority_order": ["nn_trap", "two_cluster_bottleneck", "corridor", "clustered", "grid_like", "default"],
"rules": {"clustered": {"condition": "clustering score and cluster separation", "heuristic": "cheapest_insertion_two_opt",
"op": ">=", "threshold": 0.52}, "corridor": {"condition": "corridor score / elongated structure", "heuristic":
"limited_three_opt", "op": ">=", "threshold": 0.46}, "default": {"heuristic": "farthest_insertion_two_opt"}, "grid_like":
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"two_cluster_bottleneck": {"condition": "bottleneck / two-cluster score", "heuristic": "farthest_insertion_two_opt", "op":
">=", "threshold": 0.55}}}, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 3,
"temperature_scale": 1.15, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic":
"farthest_insertion_two_opt"}

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phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.086887, selector regret 0.014627, runtime-adjusted gap 0.14749.
- Family transfer 0.000388 and combined transfer 0.032256.
- Runtime 2831.301143 ms, runtime inflation 0.697496, mean novelty 0.22521, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:edge_preserving_restart:farthest_insertion_two_opt:stag5:cand-1:restart1.
- Controller rule summary: {"acceptance_bias": 0.008, "candidate_limit_offset": -1, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selection with conservative switching tuned for transfer under diversity-failure replay mode.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "cheapest_insertion_two_opt", "low_improvement_threshold": 0.12, "name": "interpretable_instance_adaptive_tsp_controller_v3", "nearest_neighbor_trap_heuristic": "candidate_pruned_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "farthest_insertion_two_opt", "stagnation_threshold": 5, "temperature_scale": 1.0, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.105441, selector regret 0.033181, runtime-adjusted gap 0.105441.
- Family transfer 0.001007 and combined transfer 0.039482.
- Runtime 390.821043 ms, runtime inflation -0.765685, mean novelty 0.854268, and complexity 0.76.
- Pareto efficient: True.

Judge Appendix

Summary of TSP Adaptive Heuristic Portfolio Suite Evaluation

Condition Name	Held-out TSPLIB Gap	Selector Regret vs Oracle	Runtime-Adjusted Gap	Runtime Inflation	Transfer Gap	Pareto Efficient	Notes on Interpretability & Transfer
phase8_best_single_fixed_heuristic (Farthest Insertion 2-opt)	**0.07226**	0.0	0.07226	0.0	0.03122	Yes	Strong baseline; top TSPLIB gap, no runtime penalty, zero selector regret; simple and interpretable, static heuristic.
phase8_oracle_selector	0.07226	0.0	0.07270	0.006	0.02668	No	Oracle baseline (upper bound) matches best fixed heuristic in TSPLIB gap and selector regret; small runtime increase; non-deployable.
phase8_supervised_ml_selector	0.07226	0.0	0.07285	0.008	0.03122	No	Matches best fixed heuristic in TSPLIB gap and regret; slight runtime inflation; adaptive but no clear gain over fixed.
phase8_llm_static_selector	0.22927	0.157	0.22927	-0.845	0.112	Yes	Lower runtime but large TSPLIB gap and regret; deterministic, interpretable ruleset but underperforms strongly on gap.
phase8_llm_evolved_adaptive_controller	0.21431	0.142	0.21431	-0.741	0.095	No	Adaptive with interpretable instance-aware rules; better than static LLM selector but inferior gap to single-best; moderate runtime reduction; some interpretability.
phase8_llm_evolved_controller_diversity_failure_replay	0.08689	0.015	0.14749	0.697	0.032	No	Adaptive with diversity failure replay; improved regret, somewhat closer gap to oracle/fixed heuristic; higher runtime inflation (~70%); more complex controller.
phase8_full_solver_evolution	0.10544	0.033	0.10544	-0.766	0.039	Yes	Full solver evolution with some code novelty; intermediate gaps and regret; runtime improvement; balances complexity and performance well.
phase8_random_portfolio	0.19514	0.123	0.19514	-0.678	0.095	Yes	Random portfolio baseline; poor TSPLIB gap and regret despite low runtime; interpretable but ineffective.

Conservative Interpretation and Prioritized Conclusions

1. **Primary Endpoint (Held-out TSPLIB Gap):**

- The **best single fixed heuristic (farthest_insertion_two_opt)** achieves the lowest TSPLIB gap (~7.2%), which matches the oracle selector (upper bound).
 - None of the adaptive or static selectors improve meaningfully on this baseline.
 - Adaptive controllers trade increased complexity for modest or no TSPLIB gap improvements and generally incur runtime penalties or inflated regret.
2. **Selector Regret vs Oracle Portfolio:**
- The oracle and best fixed heuristic have zero selector regret, demonstrating this fixed heuristic's optimality in this setting.
 - Adaptive controllers present non-negligible regret (0.0146–0.157), indicating missed gains from oracle selections.
3. **Runtime-Adjusted Gap & Runtime Inflation:**
- Adaptive controllers reduce runtime (up to ~70% less than fixed) but increase gap substantially (>0.14).
 - The best fixed heuristic runs longer (~1.6s) but yields the best quality gap and zero regret, balancing quality and time.
 - The diversity failure replay controller trades runtime inflation (~70% higher) for reduced regret but still underperforms the fixed heuristic in absolute gap.
4. **Pareto Efficiency:**
- The best fixed heuristic, full solver evolution, LLM static selector, and random portfolio lie on the Pareto front.
 - Adaptive solutions (evolved controllers) are not Pareto efficient due to worse trade-offs between gap and runtime.
 - Pareto fronts highlight that the simple fixed heuristic is most competitive overall.
5. **Cross-Family Transfer:**
- Transfer gaps for the best fixed heuristic and oracle remain low (~0.03 or less), indicating robust generalization.
 - Adaptive controllers exhibit higher transfer gaps (up to ~0.11), signifying some generalization weaknesses under cross-family conditions.
6. **Interpretability and Instance-Adaptive Control:**
- Adaptive controllers with interpretable rules (phase8_llm_evolved_* controllers) provide instance-adaptive decisions but do not substantially close the gap to the oracle or best fixed heuristic.
 - Runtime savings are possible at the cost of solution quality and increased complexity.
 - Static selectors with deterministic heuristic preferences (LLM static selector) show poor gap performance but high interpretability and runtime efficiency.
 - No new heuristics discovered: all methods control known heuristics.

Final Recommendation

- Retain or prioritize the **best single fixed heuristic (farthest_insertion_two_opt)** as a strong and interpretable baseline for deployment due to its best held-out TSPLIB gap, zero selector regret, and balanced runtime.
- Adaptive and complex controllers provide proof-of-concept for interpretable instance-adaptive selection but currently fail to improve beyond the fixed heuristic baseline in terms of final gap or regret without runtime penalty.

- Further tuning or innovation is needed to achieve meaningful TSPLIB gap improvement with adaptive control without excessive runtime inflation.
- Oracle selector remains a valuable benchmark but is not deployable.
- Focus next phase on improving adaptive selector quality and transfer robustness, especially reducing selector regret and transfer gap while preserving runtime efficiency.